What Is .NET ?

By – Achyut V Kendre

Today’s Learning

- What is Microsoft.NET?
- Why Microsoft. Net?
- What we can do with Microsoft. Net?
- Frameworks in Microsoft.NET?
- Introduction to .NET Core.
- Why .NET Core?
- CLI Commands.
- Console Program Demo.

What is Microsoft. Net?

- Microsoft.NET is an open-source platform for building desktop, web, and mobile applications that can run natively on multiple operating system.

- The .NET system includes tools, libraries, and languages that support modern, scalable, and high-performance software development.
- An active developer community maintains and supports the .NET platform.

Why Microsoft.NET?

- Cross Platform.
- Multi Language Platform.
- One Consistent API & Libraries
- Open Source & Productive.
- Extensive Tools.
- Any app, Any Platform.
- Trusted and Performance.
- Object Oriented and Service Oriented.

What we can do?

Web

Build web apps and services for Windows, Linux, macOS, and Docker.

Mobile

Use a single codebase to build native mobile apps for iOS, Android, and more.

Desktop

Create native apps for Windows and macOS or build apps that run anywhere with web technologies.

Microservices

Create independently deployable microservices that run on Docker containers.

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What we can do?

Cloud

Consume existing cloud services, or create and deploy your own.

Machine learning

Add vision algorithms, speech processing, predictive models, and more to your apps.

Game development

Develop 2D and 3D games for the most popular desktops, phones, and consoles.

Internet of Things

Make IoT apps, with native support for the Raspberry Pi and other single-board computers.

Frameworks in .NET

Currently .NET Platform has following popular frameworks are –

- .Net Framework
- Mono.Net
- .Net Core
- Xamrin.Net

Microsoft. Net Framework

First Framework Released in 2002 with .NET Platform has following features -

- General Purpose Framework
- Object Oriented and Service Oriented.
- Multi Language Platform.
- Single Language Common.
- Common Language Runtime.
- Code Based Security – many more ...

Microsoft. Net Framework

Beside these features it also has few limitations -

- Closed Source
- Works only on Windows Platform
- Need Visual Studio
- Processor and OS architecture dependent.

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Mono.NET

- Mono is a software platform designed to allow developers to easily create cross platform applications.
- Sponsored by Microsoft, Mono is an open source implementation of Microsoft's .NET Framework as part of the .NET Foundation and based on the ECMA standards for C# and the Common Language Runtime.

mono™

.NET Core

- .NET Core is a free, cross-platform, open source application development platform for building many kinds of applications.
- .NET core is built on a high-performance runtime that is used in production by many high-scale apps. Few features -
- General Purpose Framework
- Cross Platform
- Open Source
- CLI
- Side by Side Installation

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Xamrin.Net

- Xamarin is a free and open source mobile app platform for building native and high-performance iOS, Android, tvOS, watchOS using C# ...
- Xamarin is an open-source platform for building modern and performant applications for iOS, Android, and Windows with .NET.
- Xamarin runs in a managed environment that provides conveniences such as memory allocation and garbage collection.

.NET CORE

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1. Introduction

What .NET Core?

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What .NET Core?

- .NET Core is an open-source, general-purpose development platform. You can create .NET Core apps for Windows, macOS, and Linux for x64, x86, ARM32, and ARM64 processors using multiple programming languages.
- .NET Core and APIs are provided for cloud, IOT, client UI, and machine learning.
- The project is primarily developed by Microsoft and released under the MIT License.

Why .NET Core?

- Cross platform: Runs on Windows, macOS, and Linux operating systems.
- Open source: The .NET Core framework is open source, using MIT and Apache 2 licenses. .NET Core is a .NET Foundation project.
- Modern: solve today's modern needs, including being mobile friendly, build once run it everywhere, scalable, and high performance, target all kind of devices, including IoTs and gaming consoles.

Why .NET Core?

- Performance: Delivers high performance due to it's architecture to utilize hardware efficiently.
- Consistent across environments: Runs your code with the same behavior on multiple operating systems & architectures e.g. x64, x86, & ARM.
- CLI: easy-to-use command-line tools can be used for local development & continuous integration.

Why .NET Core?

- Flexible deployment: You can include .NET Core in your app or install it side-by-side (user-wide or system-wide installations).
- Compatibility: Compatible with .NET Framework and Mono APIs by using .NET Standard specification.
- Modular Architecture: .NET Core supports modular architecture approach using NuGet packages.

Languages & Editors

- The C#, Visual Basic, and F# languages can be used to write applications and libraries for .NET Core. These languages can be used in your favorite text editor or Integrated Development Environment (IDE), including:
  - Visual Studio
  - Visual Studio Code
  - Sublime Text, Vim, etc.

What we can do?

- .NET Core exposes frameworks for building any kind of app:
  - Cloud Apps with ASP.NET Core
  - Mobile Apps with Xamarin
  - IoT apps with System.Device.GPIO
  - Windows client Apps with WPF and Windows Forms
  - Machine learning ML.NET

.NET CLI Overview

- The .NET command-line interface (CLI) is a cross-platform toolchain for developing, building, running, and publishing .NET applications.
- Few of the commands are:
  - new
  - restore
  - build
  - run
  - test
  - clean

.NET CLI Overview

- The following creates new console project in the current directory with the same name as current directory.
  - ➢ dotnet new console
- The following command creates a new console project named MyConsoleApp. The -n or --name option species the name of a project.
  - ➢ dotnet new console -n MyConsoleApp

THANKS!

Any questions?

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